

MUHAMMAD ABDULLAH HASSAN

Date of birth: 01-01-2002

Current address: Lahore, Pakistan

Contact: abdullahofficial82@gmail.com

WORK EXPERIENCE

Worked as Research Associate

Feb. 2025 – Dec. 2025

- **Summary:** Completed extensive wet-lab research in molecular biology (DNA Recombinant Techniques, Protein Purification and Quantification) and currently conducting bioinformatics analysis on dengue virus genomics, including genome mapping and evolutionary classification.
- **Supervised by:** Dr. Farhan-Ul-Haque (<https://pu.edu.pk/faculty/detail/muhammad-farhan-ul-haque1>)
- **Institute:** School of Biological Sciences, University of the Punjab, Quaid-i-Azam Campus, Lahore

Worked as Lab Instructor

Dec. 2023 – Feb. 2025

- **Lab:** Biomedical Informatics and Engineering Research Laboratory (BIRL)
- **Link:** <https://birl.lums.edu.pk/muhammad-abdullah-hassan/>
- **Summary:** Built NGS analysis pipeline using Linux terminal at BIRL, LUMS (Lahore University of Management Sciences)
- **Supervised by:** Dr. Safee Ullah Chaudhary (https://lums.edu.pk/lums_employee/3846)
- **Institute:** Lahore University of Management Sciences (LUMS)

EDUCATION

MPhil. in Biological Sciences

Feb. 2025 – Present

- **Institute:** School of Biological Sciences, University of the Punjab Lahore, Pakistan (<https://pu.edu.pk/home/department/59/>)
- **Research Supervisor:** Dr. Akhtar Ali (https://pu.edu.pk/images/cv/resume_Dr-Muhammad-Akhtar-Ali45994.pdf)

Bachelors in Bioinformatics

Sep. 2019 – Aug. 2023

- **Status:** Completed with CGPA: 3.21/4.0
- **Institute:** COMSATS University Islamabad, Sahiwal Campus
- **Major subjects studied:** Linux for Biological Sciences, Microbiology and Immunology, Introduction to Bioinformatics, Biotechnology, Bioinformatics Database Development, Bioinformatics Scripting, Bioinformatics Analysis, Structural and Functional Bioinformatics, Modern Programming Languages, Statistical Methods in Biology, Introduction to Genomics and Proteomics, Bioinformatics Software Development, Biological Modeling and Simulation.

Higher Secondary School Certificate

June 2017 – Aug. 2019

- **Status:** Completed as Pre-Medical Student
- **Institute:** Divisional Public School and College Sahiwal (Chichawatni Campus)

- **Major subjects studied:** Biology, Physics, Chemistry, English

RESEARCH INTERESTS

I do have a basic understanding and am very keen on further developing my skills and looking forward to any opportunities to learn and grow in the following fields:

- Proteomics
- RNA-Seq analysis
- WGS analysis
- Comparative Genomics
- Web Application Development

RESEARCH PROJECT

- **Title:** Antiviral drug designing through CADD against Middle East Respiratory Syndrome CoronaVirus (MERS-CoV) and Respiratory Syncytial Virus (RSV) targeting Nucleocapsid protein. o **Supervised by:** Dr. Qudsia Yousafi (qudsia@cuisahiwal.edu.pk) o **Co-Supervised by:** Mr. Muhammad Saad Khan (saad.khan@cuisahiwal.edu.pk) o **Institute:** COMSATS University Islamabad (Sahiwal Campus)

Feb. 2023 – Aug. 2023

SEMESTER PROJECTS

- **Title:** Transcription and Translation through Perl language. **Purpose:** Get information about coding for biological data. **Learning:** Bioinformatics related code generation and implementation.
- **Title:** Phylogenetic tree construction. **Purpose:** Evolutionary relationship among organisms. **Learning:** Learned about how evolution works by doing some analysis of phylogenetic trees through ClustalW2.
- **Title:** 3D structure prediction using Modeler. **Purpose:** 3D visualization of protein structure. **Learning:** Visualized some behavior of Protein tertiary structure.
- **Title:** Secondary structure prediction of proteins. **Purpose:** Evaluation of online tool. **Learning:** Working of SWISS-MODEL.
- **Title:** Linux operating system. **Purpose:** Installation and understanding of Linux OS. **Learning:** Learned basics of Linux commands.
- **Title:** Genetic algorithm in Artificial Intelligence. **Purpose:** Data analysis. **Learning:** Understanding of Machine learning approaches.
- **Title:** Cell designer. **Purpose:** Modeling and simulation of pathways. **Learning:** Analysis of different pathways through software Cell Designer.

ACADEMIC SKILLS

- **Bioinformatics skills:** Sequence alignment, Primer designing through tools, Nucleotide and Protein BLAST, UCSC genome browser, NCBI, UNIPROT, GENBANK, OMIM, INTERPRO, STRING, KEGG, iTASSER, SWISS-MODEL, ENSEMBLE, ChEMBL, Guidetopharmacology, Drug Bank, CMNPD, Root Image analysis using EZ-Rhizo-II, Hypocotyl measurements using ImageJ.
- **Primary laboratory skills:** Mammalian Cell Culture, Bacteria culture, PCR, Agarose gel Electrophoresis, SDS-PAGE, Protein Quantification, Protein Purification, Blood group detection, Antibiotic testing, Root and stem visualization through Light Microscope, DNA extraction.
- **Operating system:**
 - Linux
 - MAC
 - Windows
- **Programming languages:**
 - C++
 - Python
 - R

SCHOLARSHIPS & AWARDS

- Secured Lump-Sum Scholarship by COMSATS University Islamabad, Sahiwal Campus 2019-2023.
- Secured final year project funding from National Grassroots ICT Research Initiative (NGIRI) 2022/2023.

SOFT SKILLS & EXTRACURRICULAR ACTIVITIES

- Able to conduct statistical analysis under supervision
- Familiar with data science programming languages including R and Python
- Supervisor of myself
- Team player
- **Sports:** Football, Badminton, Volleyball
- **Languages:**
 - English (IELTS Certification C1 level - 7.0 Bands)
 - Urdu (Native)

REFERENCES

1. **Name:** Dr. Safee Ullah Chaudhary
Designation: Associate Professor in the Department of Life Sciences
Email: safeeullah@lums.edu.pk
University: Lahore University of Management Sciences (LUMS), Lahore
2. **Name:** Dr. Awais Ihsan
Designation: Assistant Professor in Department of Biosciences
Email: awais@cuisahiwal.edu.pk
University: COMSATS University Islamabad, Sahiwal Campus

3. Name: Dr. Farhan-Ul-Haque

Designation: Assistant Professor in the School of Biological Sciences

Email: mfarhan.sbs@pu.edu.pk

University: University of the Punjab, Lahore